PROJECT PRESENTATION

ONLINE COURSE MARKET PLACE

A web-based platform to discover, purchase and learn from online courses

Kalasalingam University

Team Project

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Project Team

Team details provided for the presentation

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Abstract

Project overview

Overview

Online Course Market Place is a web application designed to connect learners with digital courses and instructors. Users can explore courses, review course information, add courses to a cart or wishlist, complete checkout, and access enrolled learning content. The platform also provides role-based dashboards and an AI question assistant.

Learner value

A single place to discover courses, manage learning and track progress.

Platform value

Brings course discovery, enrolment, payments and learning tools into one connected experience.

Problem Statement

Why an online course marketplace is useful

- Course information is often spread across different websites and platforms.
- Learners may find it difficult to compare topics, levels, ratings and prices in one place.
- Managing course selection, enrolment and learning progress across separate services can be inconvenient.
- Instructors and administrators need dedicated tools to manage courses and platform activity.

THE NEED

One connected learning marketplace

Discover • Enrol • Learn

Project Objectives

What the application aims to achieve

Easy discovery

01

Help learners browse, search and filter available courses.

Smooth enrolment

02

Support cart, checkout and course enrolment workflows.

Learning access

03

Provide enrolled-course access and progress updates.

Role-based tools

04

Offer separate experiences for students, instructors and administrators.

AI assistance

05

Allow users to ask learning-related questions through an AI assistant.

Existing Approach vs Proposed System

High-level comparison

Typical fragmented approach

- Course information may be distributed across multiple sites
- Learners may manage discovery and progress separately
- Instructor and admin workflows may use different tools
- Help may depend on external support resources

Proposed Online Course Market Place

- Course browsing and search in one platform
- Cart, wishlist and checkout flow
- Student learning dashboard and progress tracking
- Instructor and admin dashboards
- Integrated AI question assistant

System Architecture

Conceptual architecture based on the submitted source code

USER INTERFACE

React 19 • Vite • React Router

APPLICATION API

Node.js • Express • REST API • Authentication & validation

DATA & INTEGRATIONS

Supabase, MongoDB • Razorpay • Google Gemini API

Frontend sends requests to backend APIs; backend services communicate with data storage and external integrations.

Major Application Modules

Functional areas visible in the project structure

Authentication

Register, login, profile and protected access

Course catalogue

Browse, search, filter and view course details

Cart & wishlist

Save courses and manage selections

Checkout & payment

Checkout flow with Razorpay integration

Learning & enrolment

View enrolled courses and update lesson progress

Role dashboards

Student, instructor and admin areas

AI assistant

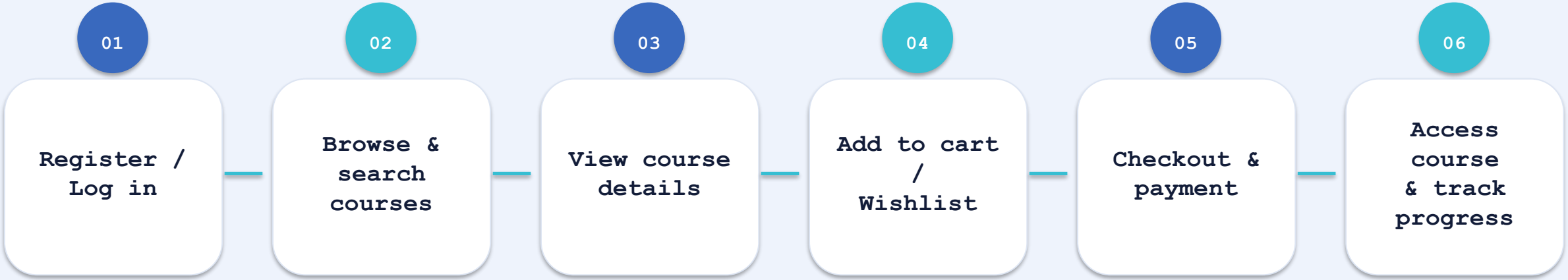
Send a question and display Gemini-generated response

Reviews

Course ratings and review-related functionality

Typical Learner Workflow

End-to-end journey through the marketplace



After successful enrolment, the learner can open learning content and update lesson completion/progress.

Technology Stack

Technologies and their roles in the application

Frontend — **React 19, Vite, React Router**

Builds the interactive user interface and page navigation.

Backend — **Node.js, Express.js**

Provides REST endpoints and handles application workflows.

Database / services — **Supabase, MongoDB**

Provides the configured backend data and authentication services.

Payments — **Razorpay**

Payment gateway integration for the checkout flow.

AI integration — **Google Gemini API**

Generates responses to user questions through the backend.

HTTP communication — **Axios**

Sends requests between the frontend and backend.

Role-Based Dashboards

Different capabilities for different platform users

Student Dashboard

- View learning activity
- Access enrolled courses
- Track course progress
- Manage profile and saved courses

Instructor Dashboard

- Dedicated instructor area
- Course-related management
- Access instructor workflows

Admin Dashboard

- Dedicated administration area
- Platform management workflows
- Administrative overview

Available dashboard pages and supporting controllers are present in the submitted project.

AI-Powered Question Assistant

Gemini integration identified in the backend source

How it works

1. User enters a question in the application.

2. Frontend sends the question to the backend AI endpoint.

3. Backend calls Google Gemini (gemini-2.5-flash).

4. The generated response is returned to the application and shown to the user.



Implementation & Demonstration Plan

Suggested items to show during the live project demo

1. **User access** — Register or log in and open the relevant dashboard.
2. **Course discovery** — Search, filter and open a course details page.
3. **Selection** — Add a course to the cart or wishlist.
4. **Checkout** — Demonstrate the checkout and payment integration in test mode.
5. **Learning** — Open an enrolled course and update lesson progress.
6. **AI assistant** — Submit a sample question and show the generated answer.

Checkout, Payments & Access Control

Supporting integrations and backend safeguards

Checkout and payments

- Cart and checkout pages support the purchase journey.
- Razorpay is configured as the payment gateway integration.
- Payment-related routes are included in the backend.

Application safeguards

- Authentication middleware protects selected API routes.
- Request validation is included in the backend.
- Central error-handling middleware helps manage API errors.
- Keep API keys and secrets in environment variables.

Conclusion & Future Scope

Summary

Conclusion

The Online Course Market Place brings course discovery, enrolment, payment and learning progress into a single web application. Its React frontend, Express backend and connected services support separate learner, instructor and administrator workflows, with an AI question assistant as an additional learning aid.

Future Enhancements

- Add certificates after course completion.
- Introduce live classes and instructor-learner messaging.
- Improve recommendations using learner interests and activity.
- Add richer analytics for learners and instructors.
- Expand automated testing and accessibility checks.

THANK YOU